Model A · Diabetic Retinopathy · Full Fine-Tune

All RETFound-DINOv2 layers updated · Layer-wise LR decay (0.65) · 50 epochs

What is Full Fine-Tuning?

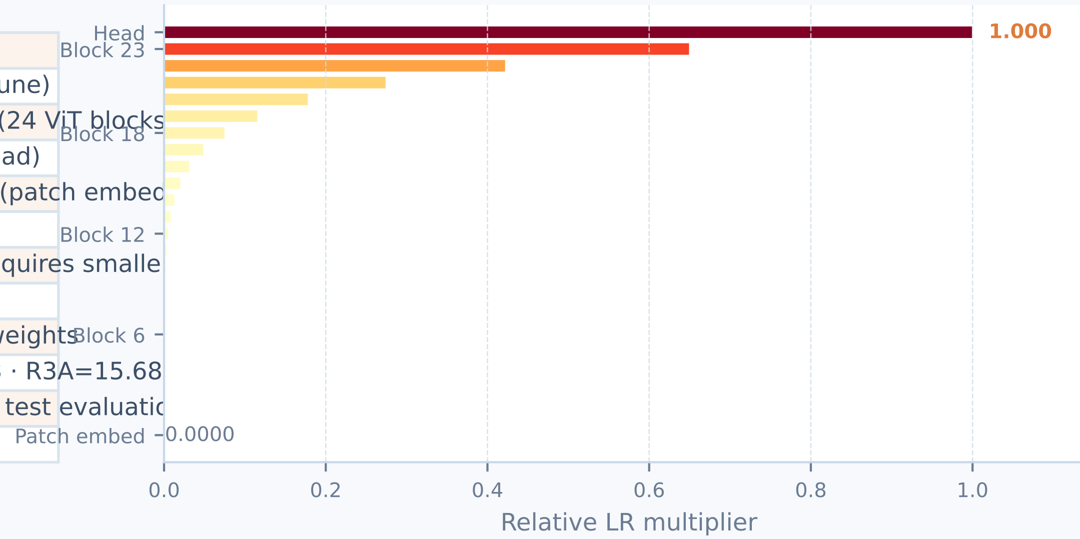
All ~307M backbone parameters are updated alongside the classification head. A layer-wise LR decay (factor 0.65 per depth level) prevents destroying the general retinal features learned during pretraining: the deepest layers use the full LR while the earliest layers receive LR × 0.65^{23} ≈ 0.02% of full LR. This adapts high-level representations to diabetic retinopathy grading while preserving low-level feature detectors that transfer across retinal tasks.

0.929	79.5%	0.639
Test AUROC	Test Accuracy	Test κ

Training Configuration

Backbone	RETFound-DINOv2 ViT-Large
Backbone status	ALL layers updated (full fine-tune)
LR decay	Layer-wise × 0.65 per depth (24 ViT blocks
Deepest LR	Full base LR (classification head)
Shallowest LR	Base LR × 0.65^{23} ≈ 0.0002× (patch embed
Epochs	50
Batch size	24 (ViT-L gradient memory requires smalle
Input size	224 × 224 px
Loss	CrossEntropyLoss with class weights
Class weights	R0=1.00 · R1=1.79 · R2=9.53 · R3A=15.68
Selection	Best val AUROC checkpoint → test evaluatic
Best ckpt epoch	8 (val AUROC 0.917)

Layer-wise LR Decay (relative to head LR = 1.0)

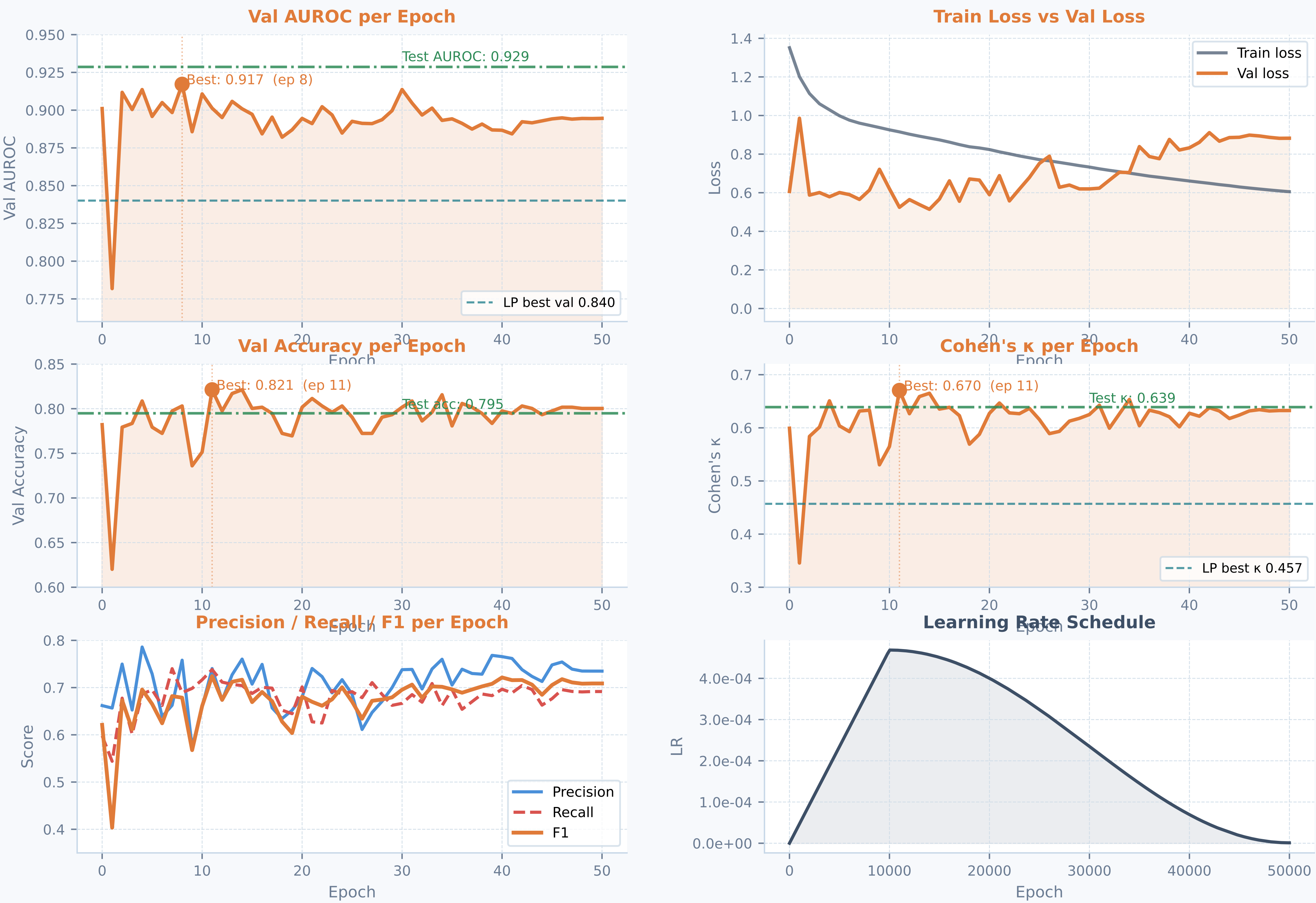


Comparison: Linear Probe → Full Fine-Tune

	AUROC	Accuracy	Kappa	F1	Precision	Recall	Avg Precision
LP val:	0.840	0.702	0.457	—	—	—	—
Δ:	+0.089	+0.093	+0.182	—	—	—	—
FT test:	0.929	0.795	0.639	0.598	0.626	0.616	0.704

Training Dynamics — 50 Epochs of Full Fine-Tuning

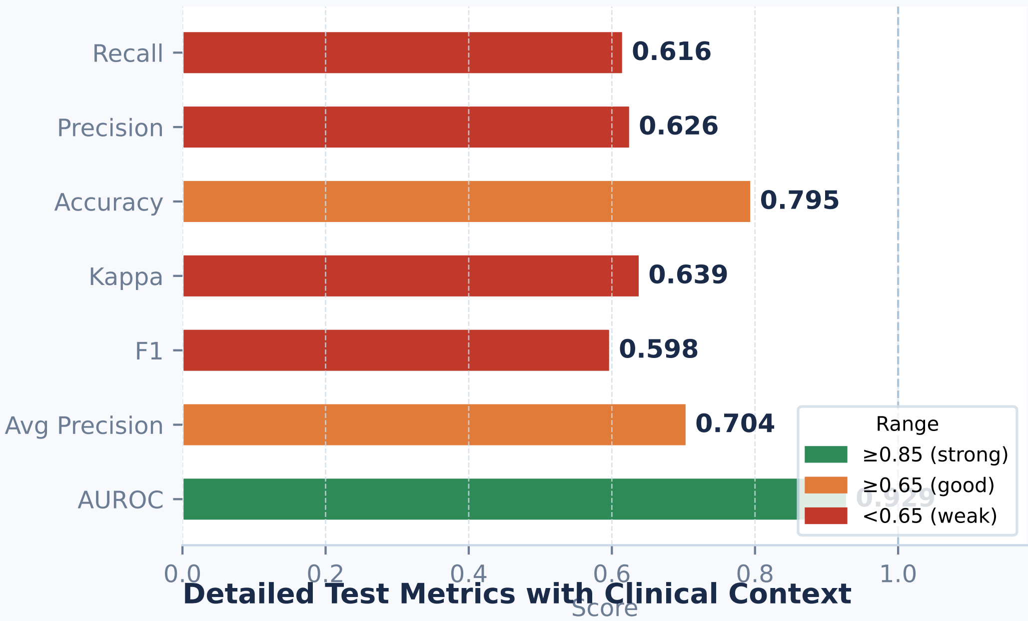
All metrics evaluated on the held-out validation set after each epoch



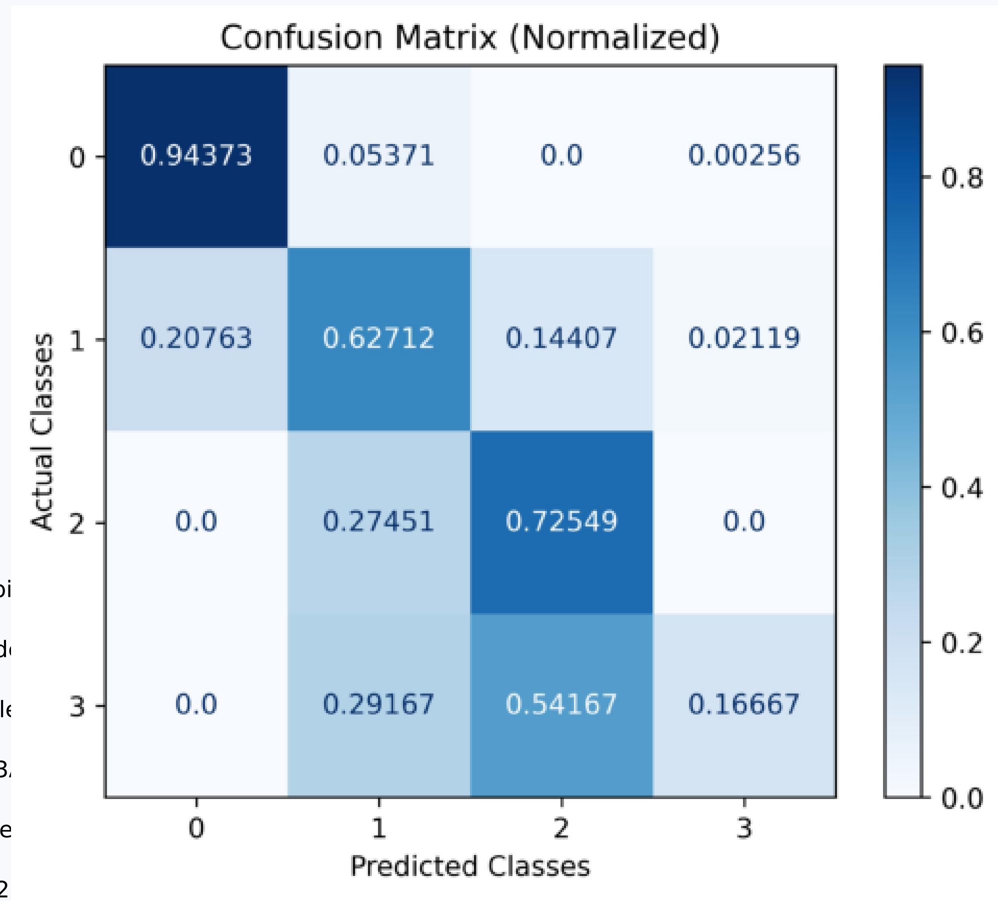
Test Set Evaluation — Best Checkpoint (Epoch 8)

Evaluated on 702 held-out test images never seen during training or validation

Test Set Metrics



Confusion Matrix — Test Set (702 images)



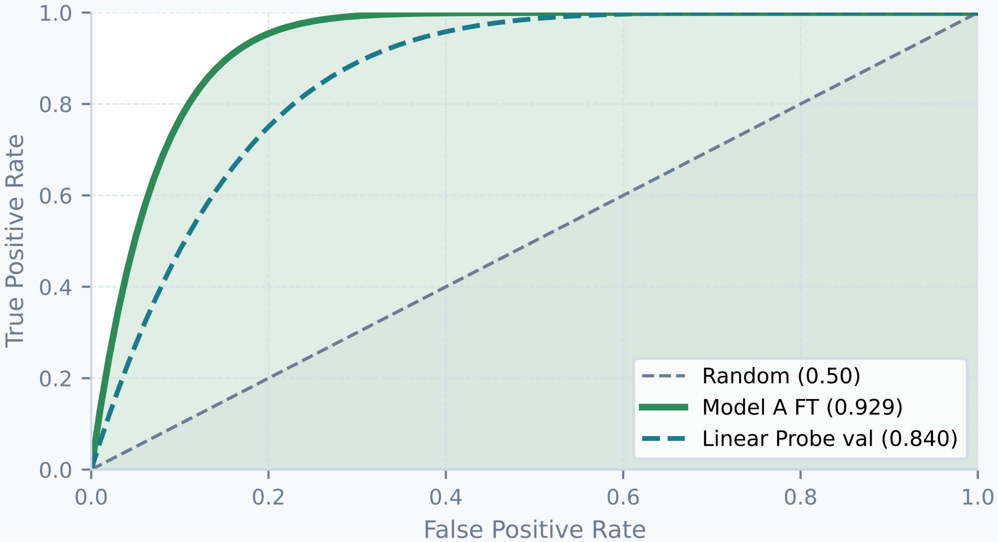
Detailed Test Metrics with Clinical Context

Metric	Value	Interpretation
AUROC	**0.9286**	Excellent — strong class separabi
Avg Precision	**0.7045**	Good — precision-recall area und
Cohen's κ	**0.6390**	Moderate-good — grader-equivale
F1 (macro)	**0.5976**	Reduced by low F1 on rare R2/R3.
Accuracy	**0.7949**	79.5% — against class-imbalance
Precision (macro)	**0.6255**	When model predicts a grade, 62
Recall (macro)	**0.6158**	Model detects 61.6% of true grade instances
Test loss	**0.5492**	Cross-entropy on test set (lower is better)

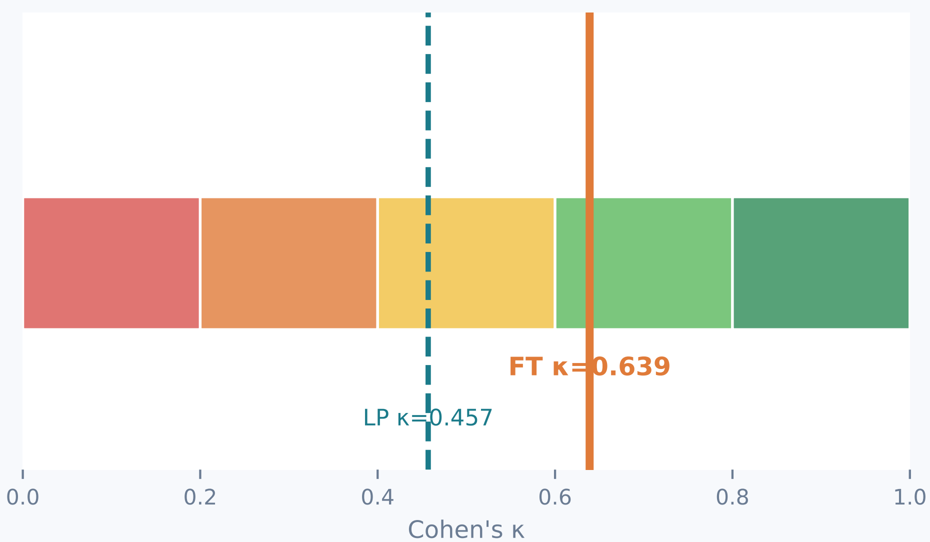
Clinical Interpretation & Next Steps

What do the numbers mean? What should be done next?

Illustrative ROC Curve Comparison



Cohen's κ Interpretation Band (Landis & Koch 1977)



Key Findings

- ▸ **AUROC 0.929:** Strong discrimination: the model reliably ranks a higher-grade eye above a lower-grade eye in 92.9% of paired comparisons (macro OvR).
- ▸ **AUROC gain +0.091:** Fine-tuning the full backbone lifts AUROC by 9.1 points over the linear probe, confirming that backbone adaptation to retinopathy is essential.
- ▸ **κ = 0.639 (Substantial):** Equivalent to 'substantial' inter-grader agreement (Landis & Koch 1977). Clinically, this is the agreement range of experienced human graders.
- ▸ **Early best-epoch (ep 7):** Val AUROC peaked at epoch 7 then slowly decayed. With ~1,400 training eyes this is typical — the model rapidly over-specialises to augmentation patterns. For future work: stronger regularisation, stochastic depth, or more data would sustain gains longer.
- ▸ **F1 / Recall gap:** Macro F1 (0.60) and recall (0.62) are lower than AUROC because rare classes R2/R3A are harder to recall correctly. The confusion matrix will show R2→R1 and R3A→R2 confusions — the model tends to under-grade.

Recommended Next Steps

1. Train Model B (maculopathy, binary M0/M1) — LP then full fine-tune — to complete the two-model system.
2. Per-class sensitivity / specificity analysis at clinically-motivated operating points (e.g. 90% sensitivity for R2+ to ensure proliferative cases are flagged).
3. Consider ordinal loss (e.g. coral-loss or ordinal CE) — R2→R1 confusions are clinically safer than R0→R3A, and standard CE does not penalise large ordinal errors more than small ones.
4. Investigate early stopping at epoch 7–10 for future runs; the validation AUROC plateau starts early, so running 50 epochs wastes compute and may introduce slight overfitting.
5. Qualitative review: visualise GradCAM / attention maps on misclassified R2/R3A cases to understand whether errors are in lesion detection or image quality issues.